

Porters	• The first beer to be brewed industrially • Blend of old and young dark brown beer • Original porters used all dark malt • Invention of the hydrometer led to switch to pale malts with burnt sugar • After the invention of the roasting kiln in 1817 switched to black malt. • <u>As stout gained popularity</u> porter became weaker paler and less bitter. • Guinness ceased production of porter in 1974.
English/American Porter	• American Porters are very similar to English porters • Dark brown colour • Medium bodied, with roasted malt aroma and flavour • Main areas of difference are due to the yeast strains and hops used. • <u>American porters</u> have low to no levels of fermentation by products (esters and diacetyl). • <u>American porters</u> generally use American hops giving a more pronounced citrusy hop aroma. • 4.8 – 6.5% ABV and 22-35 IBU
Baltic Porter	• Based on Porters exported from UK to Russia in the 18th & 19th century. • Brewed continuously in Poland Russia and the Baltic states • Modern version use lager yeast, not ale yeast • Roast/toasted malt aroma • Little hop aroma • 5.5- 9.5% ABV
Pale Ales and Bitter	• Original pale ales were brewed in London • Burton upon Trent became the predominant source of pale ales. • Burton's dominance due to the quality of its water. • High natural gypsum (calcium sulphate; CaSO_4) levels brought out the flavour in hops. • Most pale ales now brewed with liquor that has been burtonised by the addition of gypsum • In the UK these days pale ales are generally bottled • Bitter is generally on draught (on tap) and is usually dry hopped in the cask.
English Pale Ales	• Predominantly pale malts (65-100%) • Pale ale malt e.g. Maris Otter or Golden Promise • Vienna or pilsner malt • Crystal malt (1-15%) • Other e.g. wheat, chocolate, black malts, corn sugar and flaked barley • Single step infusion mash • Fermentation with English ale yeast strains • Esters

	• Diacetyl • ABV 4.5-5.5% • Crisp, balanced nutty malt and spice/herbal hop flavour • Clean or crisp finish																								
American Pale Ales	• The definitive American craft beer style • Aroma • Malty with citrus/piney American hops • Flavour • Bready/nutty/caramel malt • Grapefruit /resinous hops • Balanced hop and malt with crisp finish • Examples • Anchor Liberty Ale • Sierra Nevada Pale Ale																								
Pacific Pale Ales	•Increasing numbers of brewers are using the unique Australian and New Zealand hops. • Ella, Nelson Sauvin, Galaxy, Pacific Gem, Motueka • New & distinctive style called Pacific Pale Ale. • Examples NZPA (UK), Hapi Daze (NZ),																								
India Pale Ales	• Beers of the time were unstabilised and prone to spoilage on long sea journeys • Brewers began producing pale ales with high alcohol content and higher levels of hops • The beers had better keeping qualities. • Exported from UK by East India Company • IPA brewed for export to India and other countries. • Considered a sub style of pale ale but possibly the origin • Flavour: Toffee malts dominated by hop bitterness • Aroma: Malt and spicy/herbal hops • Crisp, dry and hoppy. <table><thead><tr><th></th><th>English Pale Ale</th><th>India Pale Ale</th></tr></thead><tbody><tr><td>OG</td><td>1.044-1.056</td><td>1.050-1.070</td></tr><tr><td>Bitterness (IBU)</td><td>20-40</td><td>40-60</td></tr><tr><td>Hop Aroma</td><td>Medium</td><td>Medium-High</td></tr><tr><td>Colour (SRM)</td><td>4-12</td><td>4-14</td></tr><tr><td>ABV</td><td>4.5-5.5</td><td>5-7.5</td></tr><tr><td>Esters</td><td>Low to med</td><td>Low to med</td></tr><tr><td>Diacetyl</td><td>Low OK</td><td>Low OK</td></tr></tbody></table> From Table 16.1 Daniels 1998		English Pale Ale	India Pale Ale	OG	1.044-1.056	1.050-1.070	Bitterness (IBU)	20-40	40-60	Hop Aroma	Medium	Medium-High	Colour (SRM)	4-12	4-14	ABV	4.5-5.5	5-7.5	Esters	Low to med	Low to med	Diacetyl	Low OK	Low OK
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India Pale Ales	• English IPAs Toffee malts dominated by hop bitterness Malt and spicy/herbal hops Crisp, dry and hoppy • American West Coast IPA Aroma dominated by American hops e.g Cascade Citra and Simcoe. • Double IPA Higher alcohol content (7.9 – 10.5% ABV) Very high bitterness (65-100 IBU) • Black IPA American style IPA with dark malt Carafo & chocolate malt • New England IPA Hazy Low bitterness, high hop aroma
Golden or Summer Ale	• A recent development from UK brewers • Essentially a lightened IPA • Golden colour • Seasonal (summer) • Flavour clean pale malt with crisp (hoppy finish) • Spicy hops (Saaz, Amarillo) • Balanced sweet: bitter or slightly hop forward (bitter) • 3.6-5% ABV • 20-28 IBU
Australian Sparkling Ale	• Recognised by BJCP guidelines as a true Australian style. • Mixture of pale and small amounts of crystal malts with Pride of Ringwood hops • Light amber colour 25-35 IBU 4.5-6.0% ABV • Persistent white head, high carbonation • Balance of malt hop ester and yeast aroma • Bready malt flavour balance by hop bitterness. • High attenuation with dry finish • Medium bodied with very high carbonation
California Common/Hybrid	• Brewed before the invention of refrigeration. • Lager yeast at ale yeast temperature (18-24°C) • Greater levels of esters than usual for a larger ale. • Generally contains predominantly pale malt with some crystal malt for flavour and colour. • Hops • Cascade for aroma • Pride of Ringwood

Scottish Ale	- Historically Scottish ales are named 60, 70, 80 and 90 Shilling (/-) depending on their strength (and price). <table><tr><th></th><th>60 Shilling</th><th>70 Shilling</th><th>80 Shilling</th><th>90 Shilling</th></tr><tr><td>Colour</td><td>16-34</td><td>20-38</td><td>30-50</td><td>50-70</td></tr><tr><td>ABV</td><td>3-4%</td><td>3.5-4%</td><td>4-4.5%</td><td>6.2-8%</td></tr><tr><td>IBU</td><td>9-20</td><td>10-19</td><td>10-19</td><td>10-49</td></tr><tr><td>Esters</td><td>0 to low</td><td>0 to low</td><td>Low to med</td><td>Medium</td></tr><tr><td>Diacetyl</td><td>Low to med</td><td>Low to med</td><td>Low to med</td><td>Med-high</td></tr></table> Daniels (1998) Table 25.1 - Hops are predominantly English Fuggles & Goldings hops. - Fermentation occurs at cooler temperatures (10-16°C) - Due to cooler fermentation the flavour profile is cleaner with lower esters and diacetyl than English bitters. - For each the flavour is predominantly nutty pale malts with hints of caramel and toffee. - Sweetness is balanced by low levels of hop bitterness. - Wee heavy is richer with greater levels of esters and diacetyl and a more complex toffee malt flavour.		60 Shilling	70 Shilling	80 Shilling	90 Shilling	Colour	16-34	20-38	30-50	50-70	ABV	3-4%	3.5-4%	4-4.5%	6.2-8%	IBU	9-20	10-19	10-19	10-49	Esters	0 to low	0 to low	Low to med	Medium	Diacetyl	Low to med	Low to med	Low to med	Med-high
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Mild & Brown Ales	- Both are made with pale base malt with crystal and dark malts for flavour and colour - Dark amber to reddish brown - Toffee, toast, nutty malt aroma and flavour with medium to low hop bitterness and aroma. - Low to moderate esters. - Medium bodied and medium carbonation <table><tr><th></th><th>60 Shilling</th><th>70 Shilling</th></tr><tr><td>ABV</td><td>3-3.8%</td><td>4.2-5.4%</td></tr><tr><td>EBC</td><td>24-50</td><td>24-44</td></tr><tr><td>IBU</td><td>20-25</td><td>20-30</td></tr></table>		60 Shilling	70 Shilling	ABV	3-3.8%	4.2-5.4%	EBC	24-50	24-44	IBU	20-25	20-30																		
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Stout	- Originally a "stout" or strong porter. - Many different substyles from weak to strong, sweet to dry. - Deep dark roasted malt character. - It has often been promoted as a healthy drink. - The addition of black patent malt in the 19th century gave stout its black colour																														

Darl

Dry or Irish Stout	• Exemplified by Guinness (other Irish stouts do exist!) • Addition of unmalted roasted barley gives it a characteristic coffee-like roastedness. • Raw unmalted flaked barley is also used a source of proteins for head retention. • Guinness was made by blending old Guinness with freshly brewed Guinness to give it a sour tang • Guinness is carbonated with a mix of N₂ and CO₂.
Foreign Extra Stout	• These stouts were primarily exported around the British Empire. • Higher strength (5.5-8.5% ABV) • Deep dark roasted malt character with some hop aroma and flavour. • Commercial examples: • Coopers Extra Stout • Nigerian and Jamaican Guinness • Southwark Old Stout
Imperial Stout	• Brewed in London in the 18th Century for export to export to the Court of Catherine the Great. • High alcohol content (8.6-14.4% ABV) • One of the few beers that can be aged
Milk Stout	• Originally promoted as nutritious and recommended for nursing mothers. • In the UK the term Milk Stout was discouraged after WWII. • Sweet roast chocolate/caramel malty flavour. • Sweetness balanced by roast bitterness, no obvious hop bitterness • 2.8 – 6% ABV • Sweetness comes from added milk sugar also called lactose. Lactose <pre> graph LR Lactose((Lactose)) -- Lactase --> Glucose((Glucose)) Lactose -- Lactase --> Galactose((Galactose)) </pre> • Glucose & galactose can be used for alcoholic fermentation. • Lactose cannot be used for alcoholic fermentation. • Yeast lack lactase activity, thus is not fermentable and remains as sweetener in the finished beer.

Oyster Stout	• In London in 1800s oysters were a traditional bar snack. • The style had all but died out. • Now resurrected as a craft beer style.. • As will other stouts dominated by roasted malt flavours and aromas. • No obvious seafood flavour.
Oatmeal & Chocolate Stout	• Oats had been a traditional addition to ales in some regions in mediaeval period. • At high levels oats can be bitter an astringent. • The use of oatmeal (flaked oat) in stout was to promote its "nourishing" qualities. • High content of proteins lipids and gums (β-glucans) add to the viscosity (body) of the drink making the mouthfeel silky. • Chocolate stouts are made with dark chocolate malt to give roasted chocolate/coffee flavour • Chocolate flavourings and other adjuncts may also be added
Old Ale	• Old English style, traditionally brewed in October and aged before consumption.. • Dark malts such as chocolate and black malt may also be added. • Traditional English hops • Low attenuation (56-70%) due to high levels on unfermentable dextrins • Malt forward, nutty caramel with light roast characteristics • Moderate to high fruity esters giving dried fruit or wine (vinous) characteristics. • Medium to full bodied, warm effect of high alcohol content. • 6-9% ABV 30-60 IBU.
Barley wine	• Originally brewed in October and matured during winter months • High gravity beer between 8-12% ABV • Amber to brown colour • Rich malty, caramel and fruity aroma • Some hop and aromatic alcohol aroma • Sweet with intense & complex malt flavours, bready, toffee/caramel flavours • Medium bodied with warming alcohol mouthfeel, possibly sherrying with aging/oxidation • American barley wines tend to have more hop aroma from the use of American hop varieties.

Red & Amber
Ales

- Modern US/Irish style
- Based on historical Irish style
- Malty aroma
- Crystal malts & roasted barley
- Slightly malt forward, low hop aroma & bitterness

Dark Ales UK; AUS

~~Aroma~~ bubble: off white bubble, 有大有小, lace, retention 有打有环

cloudness: clear body, ^{light} cloudy top

Colour: brown yellow 黄褐色 to dark red brown to black

Aroma: Dark Ale: chocolate, caramel, toffee, biscuit

Porter: earthy, ethonal, rainy, darkchoco Poland Europe

Stout: coffee shot, fruity, medicine, toasted malt

~~dry stout~~: baby powder, perfume, herbal Irish
dry stout

Taste: Dark Ale: upfront bitterness, medium-light bodied, balance,
easy to drink (bitterness disappear quickly)
coke and sour taste, ^{first} at

Baltic Porter: alcohol warming at throat finish, medium bodied.
easy to drink, moderate carbonation

Stout: dry finish, creamy, ^{→ dry stout} medium body smooth,
citrus and sweetness from start to middle.

共 36 款啤酒, under 6 categories

Pale Ales UK; USA; AUS

Aroma: 果味是 American hop, 花草或 yeast 味是 Australian ^{pale ale style} hop.

Malty, wooden 味是 English bitter / Pale Ales.

Color: range from light gold / amber to medium amber / gold colour
/ straw

bubble: ~~usual~~ good lace / foamy

Taste: hop bitterness upfront, dry finish, crisp / fresh, light-medium body

Cloudiness: mostly clear

India Pale Ales & Red and Brown Ales UK; America; AUS

Aroma: English IPAs: malty with light toffee / coffee

AUS / American IPA style: floral, lime, grapefruit hop-forward.

Red Ale: 中药味, herbal, lemon first

Scottish Ale: 冷气温, yeasty, malty (bread), 酒精味

Brown Ale: strong toffee / coffee, 炼乳

Color: Golden, copper, amber, red brown (深色系)

bubble: not foamy, good lace, but weak head retention

Cloudiness: all Very clear except some IPA

Taste: light-medium body,

English IPA: bitterness, ethanol at last
light

AUS American IPA: orange peel taste / light bitterness at last

Red Ale: dry finish, light bitterness at last

Scottish Ale: ~~card~~ cardboard at end, dry finish, sweet.

Brown Ale: fruity.